SULTAN ALI KHAN

Data Scientist & Machine Learning Engineer



Peshawar, Pakistan | sultanalikhan0344@gmail.com | linkedin.com/in/sultanalikhan149 | github.com/sultanalikhan7543 | Portfolio

RESEARCH FOCUS

01
LLM-Integrated Systems
Designing multi-tool AI platforms and analytics applications on the Anthropic Claude API - from conversational search to marketing analytics dashboards.

02
Conversational AI Architecture
Independent research on modular, context-gathering architectures for reliable, interpretable multi-domain LLM generation, in preparation for journal submission.

03
Financial ML Signal Research
Order-flow-imbalance signal modeling with LightGBM, including an independently verified synthetic pipeline validation study published as a working paper.

04
Classical ML & NLP
Real-time text classification pipelines, including a TF-IDF and Logistic Regression fake-news detector with live Streamlit inference.

05
Intelligent Sensing, Robotics & Real-World AI
AI and Machine Learning for intelligent sensing, robotics, and real-world applications, with emphasis on computer vision, deep learning, sensor-data analysis, data-driven decision-making, and intelligent systems.

EDUCATION

Bachelor of Science in Software Engineering *(2020 – 2024)*
Iqra National University, Peshawar
GPA: 3.09 / 4.0

RESEARCH & PROJECTS

Conversational Context-Gathering for Multi-Domain LLM Generation *Independent Research Paper · 2026*

- "A Modular Architecture for Reliable, Interpretable AI Systems" - unpublished manuscript, in preparation for journal submission.

Meridian - AI-Guided Platform on the Claude API *9-Tool LLM Platform*

- Began as a conversational AI search tool for scholarships, universities, jobs, research, and startups that asks clarifying questions before returning ranked results
- Expanded into a 9-tool suite: AI Search, SOP Writer, CV Builder, Study Plan Generator, Why-This-Country Writer, Thesis Writer, University Finder, Job Search, and Startup Advisor
- Built as an HTML/JS frontend calling the Anthropic Claude API
- Evaluating deployment infrastructure (Vercel/Next.js, Cloudflare Workers, Python FastAPI) to protect the API key ahead of public release

GitHub: github.com/sultanalikhan7543/Meridian

Adlytix AI - LLM-Integrated Marketing Analytics Dashboard

- Marketing analytics dashboard integrating the Anthropic Claude API

Live Demo: adlytix-ai.streamlit.app/

Flux & Flux-Sim - Order-Flow-Imbalance Signal Research *Working Paper*

- Flux: an order-flow-imbalance signal engine concept built around LightGBM
- Flux-Sim: a standalone, independently verified synthetic order-flow-imbalance pipeline validation study, decoupled from the earlier Flux concept, published as a working paper with a public GitHub repository

Research repository: github.com/sultanalikhan7543/Flux-research

Fake News Detection System *Classical ML / NLP*

- Developed a real-time fake news classifier using TF-IDF vectorization and Logistic Regression, with live text input via Streamlit
- Built a preprocessing pipeline (tokenization, punctuation removal, lowercasing) and evaluated model performance with accuracy and F1-score metrics

Repository: github.com/sultanalikhan7543/fake-news-detection

Connect - Mobile Networking App *Final Year Project*

- Designed and built a mobile application for real-time messaging, user management, and secure communication networking
- Demonstrated full SDLC implementation — from requirements gathering and UI design to testing, deployment, and documentation

PUBLICATIONS

Federated Learning for Privacy-Preserving ECG Arrhythmia Classification under Non-IID Data Distributions: A Preliminary Study

Zenodo · DOI: doi.org/10.5281/zenodo.23231472

Temporal Memory for Robust Navigation under Distribution Shift: A Controlled Benchmark of Recurrent and Memoryless Policies

Zenodo · DOI: doi.org/10.5281/zenodo.23256938

WORK EXPERIENCE

Avantduo Pvt Ltd - Operations Engineer *Melbourne, Australia · Apr 2026 – Present*

- Oversee day-to-day operational workflows across cross-functional teams, using data analytics and automation to monitor KPIs and identify process bottlenecks
- Support adoption of AI-driven tools to improve team productivity, working directly with technical and non-technical stakeholders to align execution with business objectives

Deventia Tech - Machine Learning Engineer *Jan 2025 – Jan 2026*

- Built and tested classification and regression models in Python
- Performed data preprocessing, analysis, and visualization with Pandas and NumPy on real-world datasets
- Assessed model performance against defined metrics and supported debugging, optimization, and technical documentation

UMTI Tech Solutions - AI/ML Engineer (Intern) *Feb 2026 – Mar 2026*

- Built and tested classification and regression models in Python
- Performed data preprocessing, analysis, and visualization with Pandas and NumPy on real-world datasets
- Assessed model performance against defined metrics and supported debugging, optimization, and technical documentation

TECHNICAL SKILLS

PROGRAMMING
Python, JavaScript/HTML, MQL5

AI / ML
LLM Integration (Anthropic Claude API), LightGBM, Prompt Engineering, Conversational AI System Design

TOOLS & PLATFORMS
React, Streamlit, Git/GitHub, MS Excel (VBA & reporting)

NON-TECHNICAL
Self-learning, Independent Research, Problem Solving, Adaptability

KEY NUMBERS

3.09 / 4.0	9	3	2+
GPA — B.Sc. Software Engineering	Tools in the Meridian AI Platform	Published Dev.to Articles	Years of Industry Experience

AWARDS & CERTIFICATIONS

- **Generative AI Projects** - iNeuron, Sep 2024 (text generation, prompt engineering, LLM deployment, applied AI ethics)
- **Machine Learning, AI & Data Science** — Ehunar, Dec 2024 (applied ML and statistical modelling on real datasets using Python and scikit-learn)
- **Advanced Programming in Python** - NAVTTC, Mar 2021 (one-year program covering software design patterns and applied Python development)

CONFERENCES / SEMINARS / WEBINARS

- **International Forum on Artificial Intelligence** - Islamabad, Pakistan · Feb 2026 (Physical Participation)
- **AI in Academic Research Training** - Central Library, University of Peshawar · Sep 9–10, 2026 (Physical Participation)
- **Pakistan AI Summit 2025** - Islamabad Chamber of Commerce & Industry (ICCI), Islamabad · Nov 6, 2025 (Physical Participation)

LANGUAGES

English **Urdu** **Chitrali**

REFERENCES

Name: Mr. Irfan Ullah Khan
Title: Lecturer
Relation: University Teacher (Iqra National University)
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Name: Daud Khan Khalil
Title: Assistant Professor
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